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Case Study #4: Research Project Management Office (RMPO)

16 October 2025

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Prepared by group Data Group #2



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Agenda

Project Description

Relational Model

Query

Document Diagram

Implementations

Conclusion



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Kookle University Project Initiation Form

Date: 1 Jan 2019
Project Title: Ontology-based Plant Disease Detection System
Project Description: The project aims to ...

Keywords: Ontology, Plant Disease, Intelligent System

Project Category: ☐ Research and Development
☐ Training/Capacity Building
☐ Conference/Workshop

Research Areas of the Project: Semantic Web, Ontology, Knowledge Representation

Project Duration: 12 months

Contract Period: 1 Jan 2019 – 31 Dec 2019

Budget: 1,000,000 THB

Installment:

No.	Date	%Budget	Amount
1	30 April 2019	30	300,000
2	31 August 2019	30	300,000
3	31 December 2019	40	400,000

Principle Investigator:

1. Name: Mark Zuckerberg Email: mark@kmail.com
Department ICT Department Area of Expertise: Social Analytics, AI
2. Name: Elon Musk Email: musk@kmail.com
Department ICT Department Area of Expertise: Knowledge Representation

Project Member:

1. Name: Tim Berners Lee Email: tim@kmail.com
Department ICT Department Area of Expertise: Semantic Web Ontology
2. Name: _____ Email: _____
Department _____ Area of Expertise: _____

Donor/Sponsor Name: Thailand Research Fund

Donor Type: ☐ Research Organization
☐ International Organization
☐ Private Company

Donor Contact Info: Mr. Somchai N. CMMU Building Phayathai Bangkok 10400 Thailand
Tel: (+66) 2 343 1500

Student Assistant:

1. Name: Nina Billa Email: nina@kmail.com
Department: ICT
Responsibility: Programming
Maximum working hours per month: 40 Rate per hour: 300
Start Date: 1 Feb 2019 End date: 30 June 2019
2. Name: Lucy Dell Email: lucy@kmail.com
Department: Business Management
Responsibility: Project Coordinator
Maximum working hours per month: 40 Rate per hour: 300
Start Date: 1 Jan 2019 End date: 31 Dec 2019

Main Focus of the Case Study

This case study focuses on the management of research project data at Kookle University. The RMPO (Research Project Management Office) is responsible for maintaining and tracking information about **research projects, grants, donors, and the faculty, staff, and students** involved.

OBJECTIVE

1. Maintain Data with CRUD Method
2. Analyze and report statistics for management
3. Tracking, planning and forecasting
4. Research performance of each faculty

◀ Spread sheet



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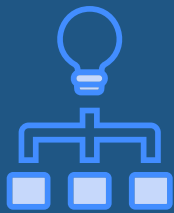
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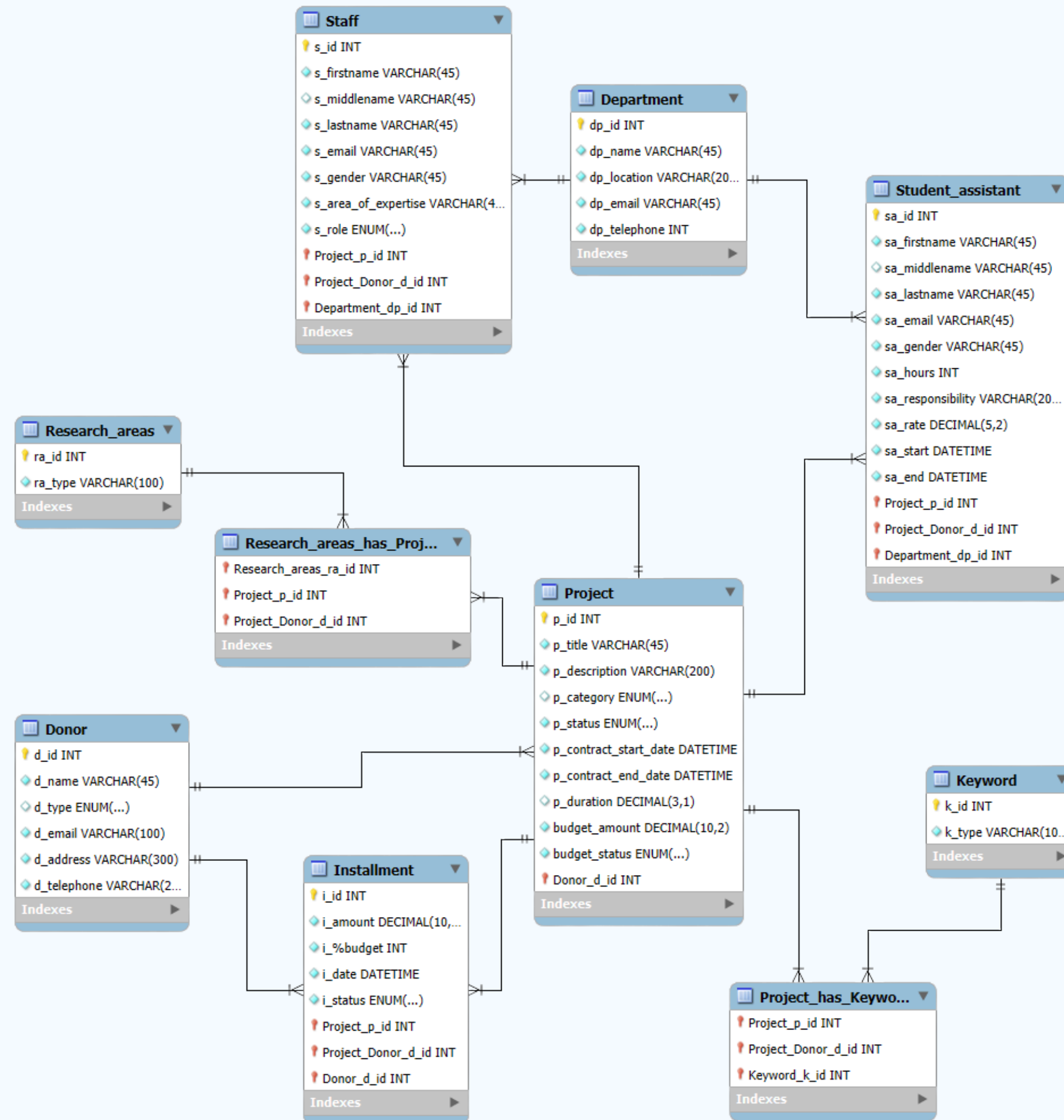
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Business Rules, Data Maintained and Constraints



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Data Maintained

- **Name:** Project Titles, User Name, Sponsor Names, Department Names, Role Names
- **Identification:** Project Code, User ID
- **Financial Data:** Budget Amount, Installment Amount, Student Assistant Rates
- **Classification:** Project Category, Donor Type, Staff Type, Research Area, Keyword
- **Date Time:** Contract Start Date, Contract End Date, Installment Date

Business Rules

- **Budget Management:** Total installments for a project must not exceed the project's total budget amount
- **Resource Allocation:** Staff and student assistants must be properly assigned to departments and projects with consistent tracking of hours and responsibilities
- **Project Classification:** Projects must be categorized as "Research and Development", "Training/Capacity Building", or "Conference/Workshop"
- **Timeline Management:** Contract end dates must be after start dates
- **Role Definition:** Staff must be classified as either "Project Member" or "Project Investigator"

Data Constraints

- **Collection :** Projects, Users
- **Embedded :** Finance, Staff, Installments, Donor, Students Assistant, Assigned_project, Department,
- **Reference :** user_id, project_code
- **Temporal Constraints:** Project_contract_start_date, Project_contract_end_date (**ISODate**)
- **Financial Values:** Project budget, Installment.amount, Student Assistant working hour rate
- **Contact Information:** Donor.contact, Department.email, Department.telephone, User.email



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Query for designing document model



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Calculate total budget amount by research area for current year

```
31 • SELECT ra.ra_type AS research_area,  
32        COUNT(p.p_id) AS project_count,  
33        SUM(p.budget_amount) AS total_budget,  
34        AVG(p.budget_amount) AS average_budget  
35 FROM Research_areas ra  
36 JOIN Research_areas_has_Project rahp ON ra.ra_id = rahp.Research_areas_ra_id  
37 JOIN Project p ON rahp.Project_p_id = p.p_id AND rahp.Project_Donor_d_id = p.Donor_d_id  
38 WHERE YEAR(p.p_contract_start_date) = YEAR(CURDATE())  
39 GROUP BY ra.ra_type  
40 ORDER BY total_budget DESC;  
41  
42
```

Identify the areas of expertise with the fewest staff members in ongoing projects, for future recruitment planning

```
SELECT s.s_area_of_expertise,  
       COUNT(s.s_id) AS staff_count  
FROM Staff s  
JOIN Project p ON s.Project_p_id = p.p_id AND s.Project_Donor_d_id = p.Donor_d_id  
WHERE p.p_status = 'On going'  
GROUP BY s.s_area_of_expertise  
ORDER BY staff_count ASC;
```

Identify top 5 research areas by total funding received

```
1 • SELECT ra.ra_type AS research_area, COUNT(p.p_id) AS total_projects, SUM(p.budget_amount) AS total_funding_allocated,  
2        SUM(COALESCE(inst_received.amount_received, 0)) AS total_funding_received,  
3        ROUND((SUM(COALESCE(inst_received.amount_received, 0)) / SUM(p.budget_amount)) * 100, 2) AS funding_efficiency_percent  
4 FROM Research_areas ra  
5 JOIN Research_areas_has_Project rahp ON ra.ra_id = rahp.Research_areas_ra_id  
6 JOIN Project p ON rahp.Project_p_id = p.p_id AND rahp.Project_Donor_d_id = p.Donor_d_id  
7 LEFT JOIN (  
8   SELECT Project_p_id, Project_Donor_d_id, SUM(CASE WHEN i_status = 'Complete' THEN i_amount ELSE 0 END) AS amount_received  
9   FROM Installment  
10  GROUP BY Project_p_id, Project_Donor_d_id  
11 ) inst_received ON p.p_id = inst_received.Project_p_id AND p.Donor_d_id = inst_received.Project_Donor_d_id  
12 GROUP BY ra.ra_type  
13 ORDER BY total_funding_received DESC  
14 LIMIT 5;
```



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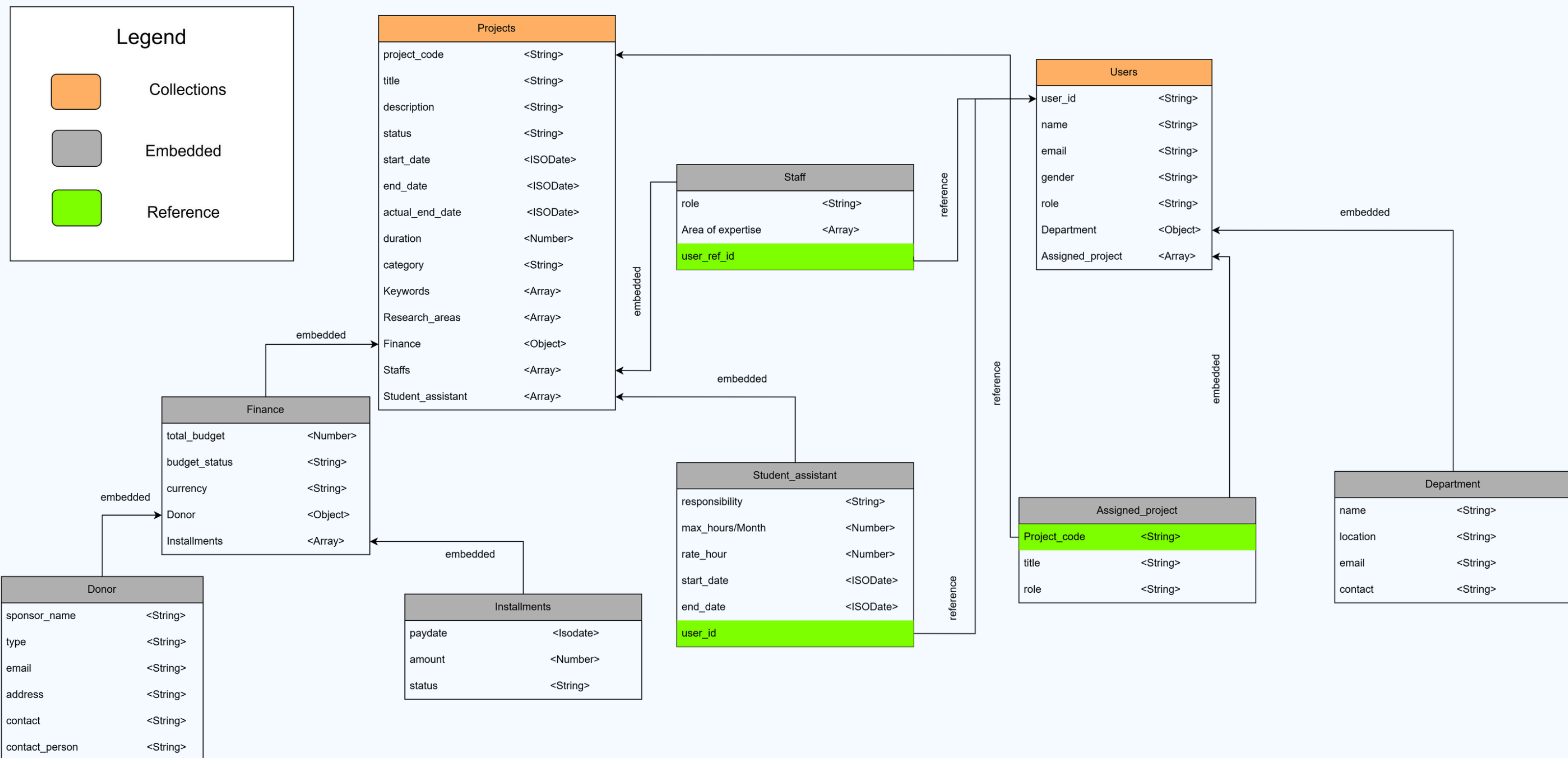
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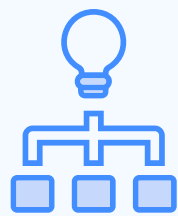
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Implementation



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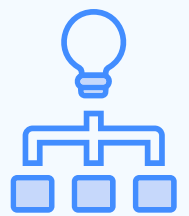
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List all projects funded by donors of type "Research Organization"

```
db.projects.find ({ "financials.donor.type":"Research Organization"},  
{_id:0,project_code:1,title:1,"financials.donor.sponsor_name": 1, "financials.donor.type" : 1 } );
```

RESULT

```
{  
  project_code: 'AI-AGRI-2025',  
  title: 'AI Application for Precision Agriculture',  
  financials: {  
    donor: {  
      sponsor_name: 'National Research Council of Thailand',  
      type: 'Research Organization'  
    }  
  }  
}
```



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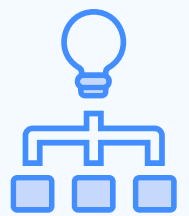
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Calculate total budget amount by research area for current year

```
db.projects.aggregate
([{$match:{start_date:{$lte:ISODate("2025-12-31T00:00:00Z")},
          end_date:{$gte:ISODate("2025-01-01T00:00:00Z")}}},
{$unwind:"$research_areas"},
{$group:{_id:"$research_areas",total_budget: { $sum:"$financials.total_budget" },
project_count:{$sum:1}}},
{$sort:{total_budget:-1}}]);
```

RESULT

```
{
  _id: 'Data Science',
  total_budget: 3000000,
  project_count: 1
}
{
  _id: 'Urban Analytics',
  total_budget: 3000000,
  project_count: 1
}
{
  _id: 'Internet of Things',
  total_budget: 3000000,
  project_count: 1
}
```



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Identify the areas of expertise with the fewest staff members in ongoing projects, for future recruitment planning

```
db.projects.aggregate([{$match:{status:{$in:["Active","On Hold"]}}},
{$unwind:"$staff"},
{$unwind:"$staff.area_of_expertise"},
{$group:{_id:"$staff.area_of_expertise",total_staff:{$sum:1}}},
{$sort:{total_staff:1}},{$limit:5},
{$project:{_id:0,area_of_expertise:"$_id",total_staff:1}}]);
```

RESULT

```
{
  total_staff: 1,
  area_of_expertise: 'Data Analytics'
}
{
  total_staff: 1,
  area_of_expertise: 'Data Science'
}
{
  total_staff: 1,
  area_of_expertise: 'AI'
}
```



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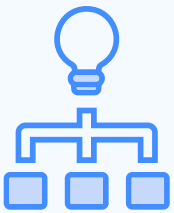
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Identify top 5 research areas by total funding received

```
db.projects.aggregate([
  {$unwind:"$research_areas" },
  {$group:{ _id:"$research_areas",total_funding: {$sum:"$financials.total_budget"}}},
  {$sort: { total_funding: -1 }},
  {$limit: 5},
  {$project: { _id: 0, research_area: "$_id",total_funding: "$total_funding"}}])
```

RESULT

```
< {
  research_area: 'AI',
  total_funding: 4000000
}
{
  research_area: 'Energy Systems',
  total_funding: 4000000
}
{
  research_area: 'Medical Technology',
  total_funding: 3500000
}
```




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Calculate total funding received from each donor type by year

```
db.projects.aggregate([
  { $unwind: "$financials.installments" },
  { $match: { "financials.installments.status": "Received" } },
  {$group: {_id:{ donor_type: "$financials.donor.type",year: { $year:"$financials.installments.pay_date" }},
    total_funding: { $sum: "$financials.installments.amount" } }},
  {$project: { _id: 0, donor_type: "$_id.donor_type", year: "$_id.year", total_funding: "$total_funding" } },
  {$sort: { year: 1, donor_type: 1} }
])
```

RESULT

```
< {
  donor_type: 'International Organization',
  year: 2022,
  total_funding: 1100000
}
{
  donor_type: 'International Organization',
  year: 2023,
  total_funding: 1100000
}
```



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Track project keyword trends over the past 5 years

```
db.projects.aggregate([
  { $match: { "start_date": { $gte: new Date("2020-10-13") } } },
  { $unwind: "$keywords" },
  { $group: { _id: { year: { $year: "$start_date" }, keyword: "$keywords" }, count: { $sum: 1 } } },
  { $group: { _id: "$_id.year", keyword_trends: {$push: {keyword: "$_id.keyword", count: "$count" }}}},
  { $sort: { "_id": 1 } },
  { $project: { _id: 0, year: "$_id", trends: "$keyword_trends" } })
```

RESULT

```
< {
  year: 2022,
  trends: [
    {
      keyword: 'big data',
      count: 1
    },
    {
      keyword: 'predictive analytics',
      count: 1
    }
  ]
}
```

```
{
  year: 2023,
  trends: [
    {
      keyword: 'surgical robot',
      count: 1
    },
    {
      keyword: 'nanoparticles',
      count: 1
    },
    {
      keyword: 'corrosion',
      count: 1
    },
    {
      keyword: 'haptics',
      count: 1
    }
  ]
}
```



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Performance for Data Retrieval

Relational Model

To assemble all the information for a single "Project Details" page, the relational model requires a complex SQL query with at least 3-4 JOINS:

```
Query 1 x
SELECT *
FROM Project P
JOIN Donor D ON P.Donor_d_id = D.d_id
JOIN Staff S ON P.Project_p_id = S.Project_p_id
JOIN Installment I ON P.p_id = I.Project_p_id
WHERE P.p_id = 1001;
```

Document Model

In contrast, with the document model we designed, all the data for the "Project Details" page is stored together in a single document . Retrieving this data is simple and fast:

```
db.projects.findOne({project_code: 'AI-AGRI-2025'})
```



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CONCLUSION

For our case study “Research Project Management Office (RMPO)” , The Document Model (NoSQL) is the more suitable choice.

The primary goal is to view, manage, and interact with projects as a whole. The document model's structure, which keeps all project-related data together in one place, is a natural and highly efficient fit for this use case. The performance benefits for data retrieval and the flexibility to evolve the application over time far outweigh the advantages of the strict normalization offered by the relational model.



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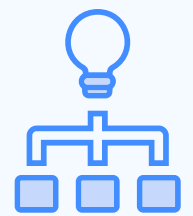
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Thank you